Nikhil Solanke

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Summary

- Built a Python–Pygame traffic simulator handling 60+ vehicles/min with YOLOv8 integration.
- Reduced junction wait time by 35% using DQN and Genetic Algorithm.
- Achieved 95% accuracy in real-time hand-gesture controlled interfaces.
- Accelerated UI prototyping by 70% using prompt engineering with no-code AI tools.

Education

Vidyavardhini's College of Engineering and Technology, Mumbai 2023 – 2027
B.E. in Computer Engineering CGPA: 7.5/10

Skills

Languages: Python, JavaScript, C++
Frameworks: ReactJS, Flask, FastAPI, OpenCV, MediaPipe, TensorFlow
Databases: MySQL, MongoDB
Expertise: Machine Learning, Computer Vision, Full Stack Development

Projects

Traffic Optimization Simulator Python, YOLOv8, DQN, FastAPI

- Simulated 4-way junction traffic and streamed live detection via FastAPI.
- Integrated YOLOv8 (92% precision) for vehicle counting and signal control.
- Reduced average wait time by 35% through DQN agent and GA tuning.

Maze Solver Robot using ESP32 C++, Arduino IDE, Embedded Systems

- Developed an autonomous robot to solve mazes using ultrasonic sensors and gyroscope-based orientation.
- Programmed in C++ to process real-time sensor data and control motors via ESP32.
- Implemented wall detection, path decision-making, and precise 90° turns using magnetometer and gyroscope.
- Hardware used: ESP32, 3 ultrasonic sensors, gyroscope, and magnetometer.

Line Following Robot C++, Arduino IDE, Embedded Systems

- Designed and built an autonomous robot capable of following black paths on white surfaces using IR sensors.
- Implemented PID control algorithm for smooth and precise line tracking.

- Programmed in C++ using Arduino IDE with real-time sensor feedback to adjust motor speeds.
- Hardware used: Arduino Uno, IR sensor array, L293D motor driver, and DC motors.

Expression Tracking System OpenCV, MediaPipe, Deep Learning

- Built facial & posture tracker at 25FPS using MediaPipe landmarks.
- Trained CNN to detect 5 emotional states with 88% accuracy.

Gesture-Controlled Interface PyAutoGUI, MediaPipe

- Implemented 4 hand gestures for mouse control (move, click, drag, swipe).
- Achieved 95% accuracy and improved user input speed by 40%.

Clubs & Activities

Core Committee Member, Yantrika (VCET) – Worked on embedded systems projects, microcontrollers, and hardware-software integration for technical events and robotics demonstrations.

Achievements

- 1st Place – Maze Solving Competition, St. John College (2024)
- 1st Place – Line Following Robot, SPIT College (2023)
- 5th Place – Line Following Robot, IIT Jodhpur TechFest (2023)